# Research Log

## Kinematic Drivers of Swing Performance: Sequencing, Plane, and Power

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### Initial Observations

*These are observations made upon initial viewing of the data and after validating with code.*

- The data sets do not all have the same columns
    - Columns not shared among all batters: `contact_x, exit_speed_mph, horizontal_launch_angle_x, vertical_launch_angle_x, hand_trigger_x`
- Data is captured at a high frequency (0.00333 seconds between frames), 900 frames per batter
- I checked the "scenario" of each take:
    - The files named batter2.csv and batter3.csv represent the same take, with batter3 having an extra column
    - Batters 1 and 5 bat left (verified by looking at hip positions along the y-axis)
    - Batter 2/3 did not make contact with the ball

### Assumptions

*These are assumptions made after examining the data and its values, using context to make assumptions.*

- Units:
    - Angular velocities are in degrees per second (°/s)
    - Speed, unless otherwise specified is in feet per second (ft/s)
    - Distances in feet (ft)
- Orientation: would affect analysis if I interpreted the dimensions as different directions (would focus on a different column)
    - y-axis is the line from home plate to the mound
        - +y direction is towards the pitcher
    - x-axis is the line "along the ground" running perpendicular to the y-axis (direction of batter's feet when in stance in the batter's box)
        - +x direction is on the first base side of the y-axis
    - z-axis is the line straight up out of the ground
        - +z direction is towards the sky
    - (0,0,0) is ground level at the back tip of home plate
- Measurement:
    - The last `NA` row before the `ball_hit_*` measurements take on numeric values is the point of contact with the ball
        - Does not always match the value in the `contact_x` column
- Motion capture subject
    - Motion capture sensors stay secure throughout swing/are tracking the same point throughout
    - Rotation around the z-axis is the primary mechanical driver of bat acceleration (towards ball)
        - Why the angular velocity analysis focuses on the z component

## Questions Considered (Ranked)

| Question | Why could it be worth pursuing? |
|---|---|
| 1. How does an efficient kinematic sequence relate to swing plane matching, and how do these jointly influence swing performance (bat speed and exit velocity)? | ● Investigates the relationship between proper sequencing and swing adjustability<br>● Connects how a batter's body generates energy "internally" with external spatial accuracy<br>● Could reveal if a batter is limited by power or bat path (since a clean sequence can still have poor plane matching) |
| 2. How does the pitch speed shift the point in the pitch's flight at which a batter commits to their swing path? | ● A physical manifestation of the mental/cognitive side of hitting<br>● Identify the "point of no return" where a batter commits to their swing path regardless of the pitch's movement<br>● Could help determine if a batter is suffering from a mechanical breakdown vs. an issue with processing |
| 3. How does the angular velocity ratio of the lead and rear wrists influence bat speed and attack angle? | ● Debate of push vs. pull mechanics<br>● Would be able to see which hand dominates in which part of the swing<br>● Could reveal force deficits that may be affecting swing mechanics |

## Prioritization

One of the reasons why the primary question was selected is because of its direct diagnostic and practical applications. It separates power generation (the sequence) from spatial delivery (swing plane matching), which can allow performance coaches to diagnose whether a batter is power-limited (inefficient sequencing) or path-limited (barrel control). Swing mechanics can then be adjusted accordingly, with drills specific to the area of most need. In addition to that, this question is better supported by the available data. A limitation faced by all three questions is the number of unique swing attempts available to analyze (4). With the scope of this question, evaluating the kinematic sequence and swing-plane is less dependent on the pitch context, speed, etc. than the other questions. The information needed to investigate this question also aligns well with the motion-capture data.

## Questions Abandoned

*How does the pitch speed shift the point in the pitch's flight at which a batter commits to their swing path?*

- Determining when a hitter cognitively decides to commit to the pitch would be most effectively investigated with eye-tracking data (gaze) and muscle activation markers. Relying on skeletal movement data alone risks mistaking movement as decision making.
- Although relevant to the selected question too, this question is more sensitive to potential confounding environmental variables. Delayed swings/commitment to swings could be caused by

other factors such as physical fatigue or just poor timing, which can't be accounted for with this data.
- Again, although all three questions are limited by the sample size, this question would be best investigated with multiple attempts by multiple batters, as well as a mix of takes, misses, and varied pitches/speeds.

*How does the angular velocity ratio of the lead and rear wrists influence bat speed and attack angle?*

- Since both hands are holding the same rigid bat handle, their movement paths are not independent of each other. With the tracking data we have, movement paths and any calculated metrics would be susceptible to collinearity.
- Although the data has the angular velocities and location data of each hand, we only have information on the physical output of the movement, not muscle activation or force application to the handle. I would not feel confident attempting to isolate the push-pull intent from the kinematic data alone.

## Unexpected Findings

- This could've been seen just from looking at the data, but after making the angular velocity vs. time graphs, I was surprised by the sudden coordinate inversions for the lead hands just prior to ball contact, for both left and right handed batters
    - 3D rotation that I didn't consider
- The batter who missed the ball had the highest peak bat speed
    - Sequence was out of order → sheer rotational velocity without control?

## Analytical Changes

- I initially plotted the raw angular velocities vs. time, but there were a lot of small peaks, making it difficult to determine which one I should be looking at
    - Applied a 4th order low-pass Butterworth filter to smooth the high frequency noise
    - Used a 13 Hz cutoff (Dowling et al., 2026)
- I replaced the indexing of the time of contact (first value in `contact_x`) with code to dynamically pick out the last `NA` row of `ball_hit_x` for batters who made contact
    - The manual marker was not reliable for establishing the start of contact
- For batters who whiffed, I used the row with the last valid approach angle data as the "time of contact"
    - Was originally using the row with their peak bat speed, but after looking at data realized the ball had already passed them

## Next Questions

- With additional and IDed data: does a batter maintain the same sequence timing lag when adjusting to off-speed pitches or boundary pitches (ex. High and inside vs.low and outside)?
- Is swing power driven more by the precision of the timing lags or the magnitude of angular velocity?
- Does fatigue (within game or seasonal) alter sequencing lags, peak velocities, and swing-plane alignment? Which part of the sequence is the first to go?
- How does off-timing in different parts of the kinematic sequence affect bat speed? Does one segment have more of an impact than another?

## Analyses Not Conducted

- Formal statistics and regression modeling, testing statistical significance of sequencing lags on exit velocity
    - With the small sample size, any models would lack statistical power which could lead to overfitting and misleading conclusions
    - This analysis is treated more as a descriptive case study, looking at the individual sequences and seeing what could be inferred on this group based on the evidence we do have
- Did not consider pitch locations and speeds as a factor in affecting the timing lags and sequencing
    - Evaluating spatial adjustment requires a large, multi-pitch dataset spanning different strike-zone areas
    - This dataset was bound to four swings without specified strike zones
    - Traded context analysis for a focus on general sequence efficiency and plane matching
- Evaluating the repeatability or trend of timing lags across multiple swings for a single batter
    - We could not assume that any of the batters were the same person
    - Traded individual player consistency analysis for more of a comparative profile

## References

Dowling, B., Slowik, J. S., & Fleisig, G. S. (2026). The influence of sample and filter frequencies on baseball pitching data. Sports biomechanics, 25(7), 971–992. https://doi.org/10.1080/14763141.2026.2625797